

Vittoria De Pellegrini

📍 Thuwal, Saudi Arabia — 📞 +966 56 201 9451 — ✉ vittoria.depellegrini@kaust.edu.sa — 🔗 linkedin.com/in/vittoria-de-pellegrini-110371229
— 🌐 github.com/VittoDePe98 — 🏠 gitlab.com/VittoDePe98 — 🐳 hub.docker.com/u/vittodepe98 — 🎓 scholar.google.com — ☰
deepwave.kaust.edu.sa/people

Professional Summary

Italian-Brazilian PhD student at King Abdullah University of Science and Technology (KAUST) and Doctoral Researcher with the industry-funded DeepWave Consortium, an initiative that builds machine- and deep-learning workflows for subsurface reservoir characterization. I focus on deep generative AI—especially video diffusion models—to model multi-phase subsurface fluid-flow dynamics. My workflows rely on CUDA-aware, reproducible CPU/GPU parallel computing pipelines running at scale on the KAUST IBEX Supercomputing cluster.

Skills

Programming & Tooling: Python, C++, Visual Studio Code, Git/GitHub, Docker/Docker Hub, Linux terminal workflow

Machine/Deep Learning & Generative AI: Supervised, semi-supervised, and unsupervised learning; linear/logistic regression; random forest; gradient boosting; image segmentation; neural networks; fine-tuning workflows; diffusion models (conditional, latent, video); progressive autoregressive video diffusion models; diffusion transformers; latent space engineering; VAE/VQ-VAE/VQ-GAN (2D/3D)

Python Libraries: PyTorch, TensorFlow, Hugging Face Transformers, Accelerate, Optuna, TensorBoard, Weights & Biases (wandb), NumPy, Pandas, Scikit-Learn, scikit-image, Matplotlib, Seaborn, Folium, Dlisio, Lasio, OPM Flow, pyopmspe11 (SPE11 benchmark), Pickle, JSON, QB Styles

High-Performance Computing: CPU/GPU parallel programming, CUDA job scripting, KAUST IBEX Supercomputing

Productivity & Collaboration: Microsoft Word, Excel, PowerPoint, Overleaf, LaTeX, Slack, Zoom

Professional Experience

DeepWave Research Consortium, KAUST — Doctoral Researcher

Jan 2024 – Present

Thuwal, Saudi Arabia

- Build deep generative models for multi-phase subsurface fluid-flow dynamics.
- Build latent conditional diffusion models and latent progressive autoregressive video diffusion models tailored to reservoir simulation data.
- Generate video sequences of key field variables such as CO₂ gas saturation and pressure build-up evolution in the subsurface for CO₂ geological sequestration studies.
- Run large-scale training and inference experiments on the KAUST IBEX multi-GPU cluster through CUDA-aware job scripts and CPU/GPU parallel workflows.

Politecnico di Torino (Polito) — Graduate Researcher

Feb 2023 – Jul 2023

Turin, Italy

- Developed supervised machine learning models for conventional and advanced well-log prediction.
- Applied the models to wells from the São Francisco and Santos basins in Brazil using PETROBRAS datasets.

Aramco Overseas Company B.V. — Geophysics Intern

Sep 2022 – Nov 2022

Delft, Netherlands

- Built a 3D multi-physics synthetic model for a geothermal site in Saudi Arabia.
- Generated surface wave dispersion curves.
- Produced synthetic seismograms.

Education

King Abdullah University of Science and Technology (KAUST)

Jan 2024 – Dec 2027 (Expected)

Doctor of Philosophy in Earth Science and Engineering — GPA 3.89

Thuwal, Saudi Arabia

Politecnico di Torino (Polito)

Oct 2020 – Jul 2023

Master of Science in Petroleum Engineering — Grade 108/110

Turin, Italy

Università degli Studi di Padova (UniPD)

Oct 2017 – Jul 2020

Bachelor of Science in Civil Engineering — Grade 110/110

Padua, Italy

Liceo Scientifico Statale Giordano Bruno

Sep 2012 – Jun 2017

High School Diploma

Venice, Italy

Orewa College

Jul 2015 – Sep 2015

Exchange Summer Student

Auckland, New Zealand

Languages

Italian — Bilingual • Brazilian Portuguese — Bilingual • English — Advanced • Spanish — Intermediate • Khaliji Arabic — Basic

Conference Publications

V. De Pellegrini, D. Wamriew, T. Alkhalifah. (2025, June) “A Conditional Diffusion Model for CO₂ Monitoring and Forecasting in Heterogeneous Geological Formations.” *86th EAGE Annual Conference & Exhibition* (Vol. 2025, pp. 1 - 5). European Association of Geoscientists & Engineers.

V. De Pellegrini, T. Alkhalifah. (2025, August) “Towards Generative Modeling of CO₂ Geological Storage with Latent Conditional Diffusion Models.” *Fifth International Meeting for Applied Geoscience & Energy*. SEG/AAPG, Expanded Abstracts.

Thesis

V. De Pellegrini. “Development of Supervised Machine Learning Models for the Prediction of Well-Logs & Application on Wells at São Francisco and Santos Basins, Brazil.” Master’s Thesis, Politecnico di Torino, 2023.